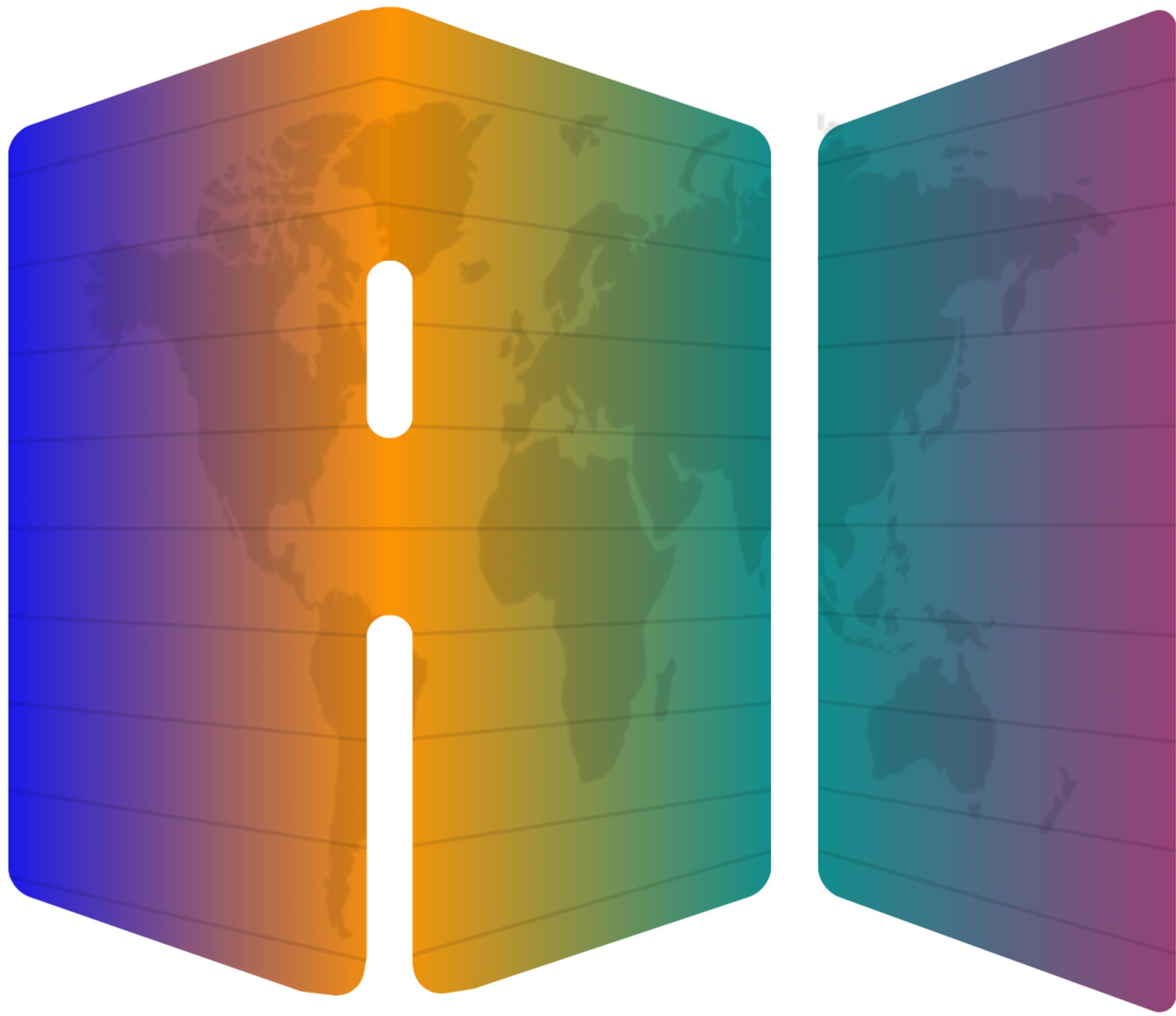
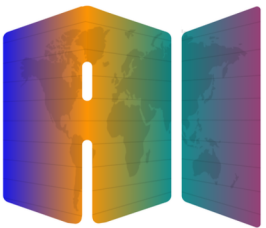




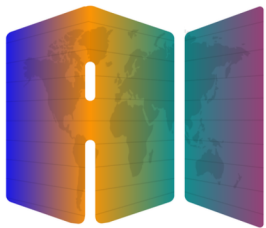
Agentic AI

Lectures by Andrew Ng



Patterns for highly autonomous agents

Planning workflows



Planning example: Customer service agent

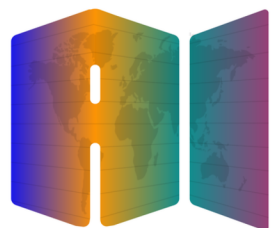
Inventory Database

id	name	description	price	stock
1001	Aviator	Timeless pilot style for any occasion, metal frame	4920	12
1002	Catseye	Glamorous 1950s profile plastic frame	3690	28
1003	Moon	Oversized round style, plastic frame	7380	15
1004	Classic	Classic round profile, gold frame	3690	9

Customer query:

Do you have any
round sunglasses
in stock that are
under Php6000?

Yes we have our **Classic** sunglasses, which are a classic round metal frame and cost Php3690



Planning example: Customer service agent

system
prompt

You have access to the following tools:

{description of tools}

Return a step-by-step plan to carry out
the user's request

Tools

get_item_
descriptions

check_
inventory

process_
item_return

get_item_
price

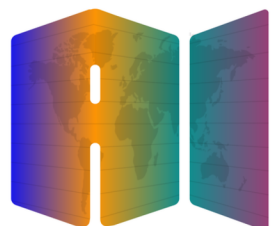
check_past_
transactions

process_
item_sale

Do you have any
round sunglasses
in stock that are
under Php6000?

LLM

1. Use **get_item_descriptions** tool to find round sunglasses
2. Use **check_inventory** to see if results are in stock
3. Use **get_item_price** to see if in-stock results are <Php6000



Step 1 text

LLM

get_item_
descriptions

Step 1 output
Step 2 text

LLM

check_
inventory

Step 2 output
Step 3 text

LLM

get_item_
price

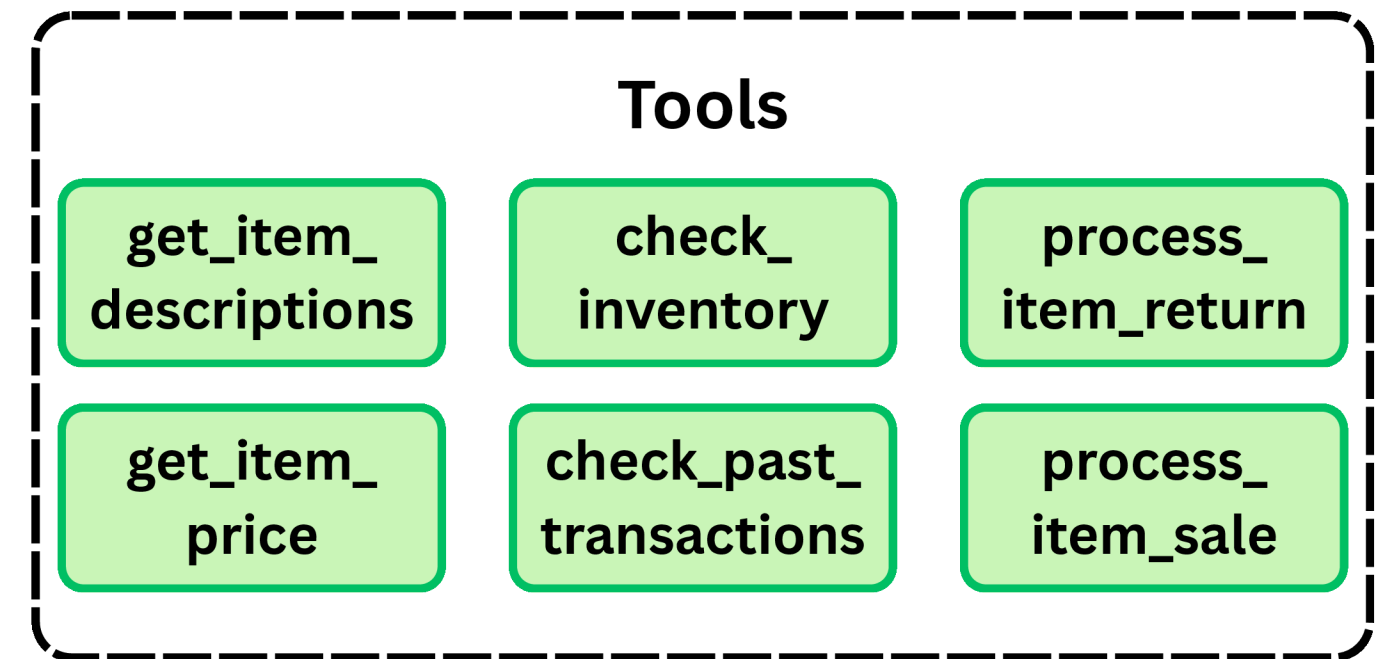
Planning example: Customer service agent

system
prompt

You have access to the following tools:

{description of tools}

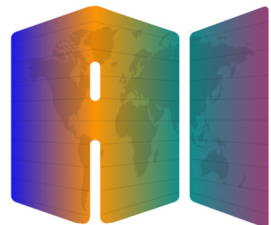
Return a step-by-step plan to carry out
the user's request



I would like to
return the gold
frame glasses I
purchased, but not
the metal frame
ones.

LLM

1. Use **check_past_transactions** to find
which glasses they bought
2. Use **get_item_descriptions** to find
the gold frame glasses
3. Use **process_item_return** to return
the gold-framed glasses



Step 1 text

LLM

check_past_transactions

Step 1 output
Step 2 text

LLM

get_item_descriptions

Step 2 output
Step 3 text

LLM

process_item_return

Planning example: Email assistant

system
prompt

You have access to the following tools:

{description of tools}

Return a step-by-step plan to carry out
the user's request

Tools

search_
email

move_
email

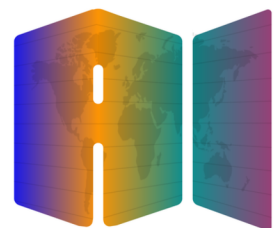
delete_
email

send_
email

Reply to that email
invitation from
Bob about dinner
in Makati and tell
him I'll attend.
Then archive his
email

LLM

1. Use **search_email** to find emails from "Bob" that mention "dinner" and "Makati"
2. Use **send_email** tool to reply and confirm attendance
3. Use **move_email** tool to move email to "archive" folder



Step 1 text

LLM

search_
email

Step 1 output
Step 2 text

LLM

send_
email

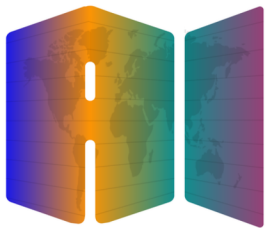
Step 2 output
Step 3 text

LLM

move_
email

Patterns for highly autonomous agents

**Creating and executing
LLM plans**



Planning example: Customer service agent

system
prompt

You have access to the following tools:

{description of tools}

Return a step-by-step plan to carry out
the user's request

Tools

get_item_
descriptions

check_
inventory

process_
item_return

get_item_
price

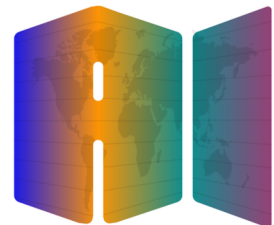
check_past_
transactions

process_
item_sale

Do you have any
round sunglasses
in stock that are
under Php6000?

LLM

1. Use **get_item_descriptions** tool to find round sunglasses
2. Use **check_inventory** to see if results are in stock
3. Use **get_item_price** to see if in-stock results are <Php6000



Step 1 text

LLM

get_item_
descriptions

Step 1 output
Step 2 text

LLM

check_
inventory

Step 2 output
Step 3 text

LLM

get_item_
price

Formatting plan as JSON

Updated system prompt

You have access to the following tools:

{description of tools}

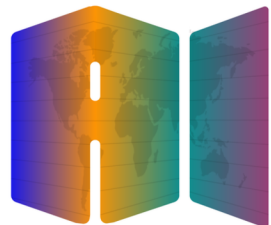
Create a step-by-step plan in JSON format.

Each step should have the following items:
step number, description, tool name, and
args.

Do you have any
round sunglasses
in stock that are
under Php6000?

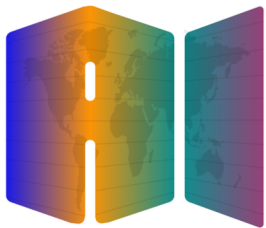
LLM

```
{  
  "plan": [  
    {  
      "step": 1,  
      "description": "Find round sunglasses",  
      "tool": "get_item_descriptions",  
      "args": {"query": "round sunglasses"}  
    },  
    {  
      "step": 2,  
      "description": "Check available stock",  
      "tool": "check_inventory",  
      "args": {"items": "results from step 1"}  
    },  
    ...  
  ]  
}
```



Patterns for highly autonomous agents

**Planning with
code execution**



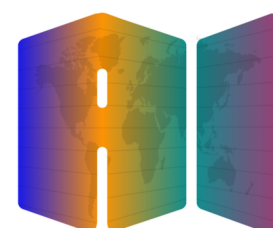
The challenge of planning with tools



Which month had
the highest sales of
hot chocolate?

LLM

date	price	coffee_name
2024-01-12	3.87	Latte
2024-01-28	2.89	Hot Chocolate
2024-02-09	3.87	Hot Chocolate
...	...	...
2025-03-23	4.57	Latte



Tools

get_column_
max

get_column_
mean

filter_rows

get_column_
min

get_column_
median

sum_rows

1. Use the **filter_rows** tool to extract transactions in January with coffee_name “Hot Chocolate”
2. Use the **get_column_mean** to find average amount
3. Use the **filter_rows** tool to extract transactions in February with coffee_name “Hot Chocolate”
4. Use the **get_column_mean** to find the average amount
5. Repeat for March, April, May, ..., December
6. Determine the month with highest average using results of previous steps

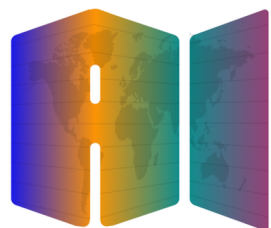
The challenge of planning with tools



Which month had the highest sales of hot chocolate?

LLM

date	price	coffee_name
2024-01-12	3.87	Latte
2024-01-28	2.89	Hot Chocolate
2024-02-09	3.87	Hot Chocolate
...	...	...
2025-03-23	4.57	Latte



Tools

get_column_max

get_column_mean

filter_rows

get_column_min

get_column_median

sum_rows

How many unique transactions last week?

get_unique_entries

What were the amounts of the last 5 transactions?

get_last_N_values

- Brittle
- Inefficient
- Continuously dealing with edge cases

...

...

Planning with code execution

system
prompt

Write code to solve the user's query. Return your answer as python code delimited with `<execute_python>` and `</execute_python>` tags.

What were the last
amounts of the last
5 transactions?

LLM

```
<execute_python>

import pandas as pd

#Load CSV
df=pd.read_csv("transactions.csv")

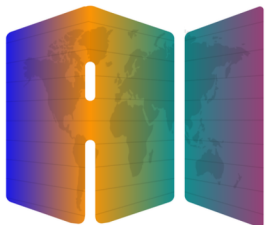
#Ensure date column is parsed as datetime
df["date"] = pd.to_datetime(df["date"])

#Sort by date to get most recent transactions
df_sorted = df.sort_values(by="date", ascending = False)

#Select the last 5 transactions
last_5 = df_sorted.head(5)

#Show just the price column (amounts)
print("Last 5 transaction amounts:")
print(last_5["price"].to_list())

</execute_python>
```



Planning with code execution

system
prompt

Write code to solve the user's query. Return your answer as python code delimited with `<execute_python>` and `</execute_python>` tags.

How many unique transactions last week?

LLM

```
<execute_python>
```

```
import pandas as pd
```

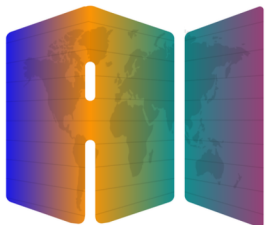
```
#Read CSV and parse the "date" column as datetime  
df = pd.read_csv("transactions.csv",  
parse_dates=["date"])
```

```
#Define time window  
today = pd.Timestamp.today()  
week_ago = today - pd.Timedelta(days=7)
```

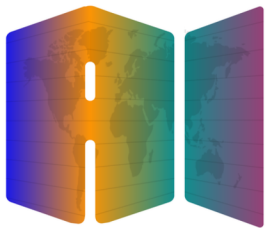
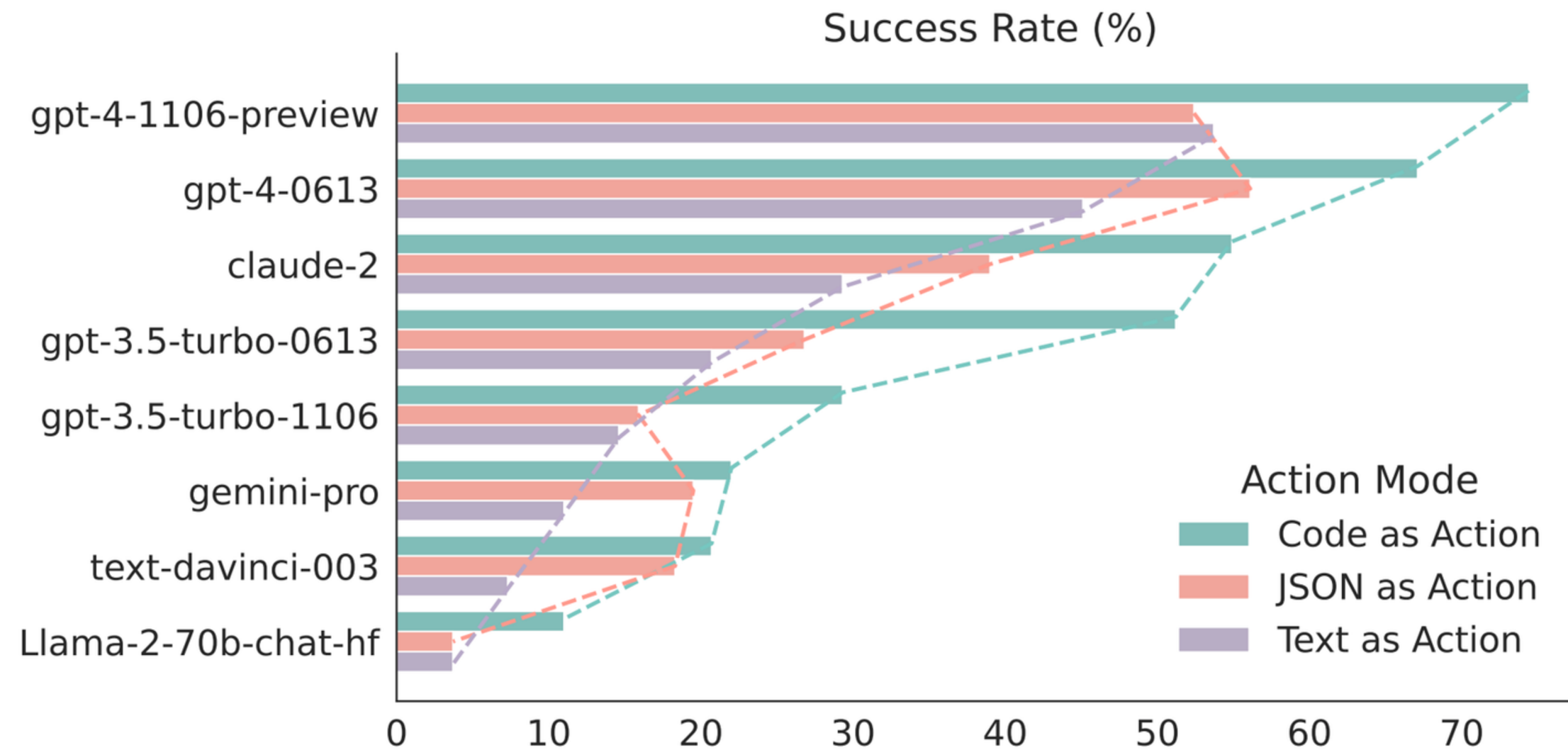
```
#Filter rows where date is within last week  
last_week = df[df["date"].between(week_ago, today)]
```

```
#Drop duplicate rows and count  
print(last_week.drop_duplicates().shape[0])
```

```
</execute_python>
```



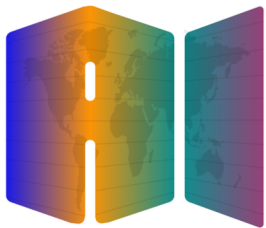
Planning with code improves performance



[Adapted from “Executable Code actions Elicit Better LLM Agents”, Wang et al. 2024]

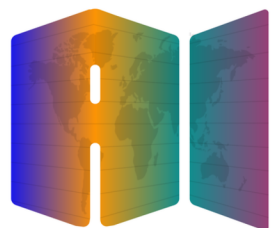
Patterns for highly autonomous agents

Multi-agentic workflows



Some tasks require more than 1 person!

Task	Team
Creating marketing assets	Researcher Graphic Designer Writer
Writing a research article	Researcher Statistician Lead writer Editor
Preparing a legal case	Associate Paralegal Investigator



Example: Marketing team

Researcher

Tasks

- Analyze market trends
- Research competitors

Tools

- Web search

researcher

Graphic designer

Tasks

- Create data visualizations
- Create artwork

Tools

- Image generation, manipulation
- Code execution for chart generation

graphic
designer

Writer

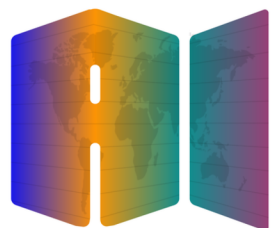
Tasks

- Transform research into report text and marketing copy

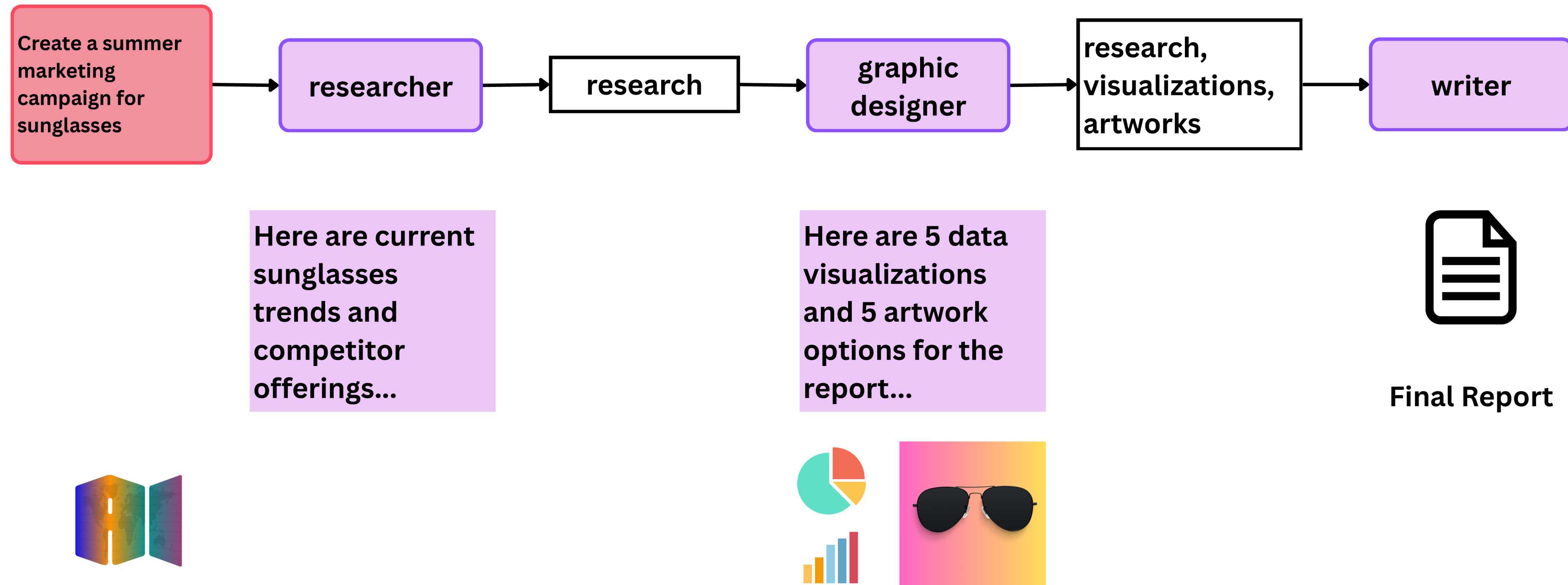
Tools

- (None)

writer



Example: Marketing team with linear plan



Example: Planning with multiple agents

system
prompt

You have access to the following tools:

{description of tools}

Return a step-by-step plan to carry out
the user's request

Create a summer
marketing
campaign for
sunglasses

LLM

Tools

get_item_
descriptions

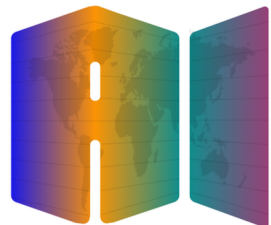
check_
inventory

process_
item_return

get_item_
price

check_past_
transactions

process_
item_sale



Example: Planning with multiple agents

system
prompt

You have access to the following tools:

{description of tools}

Return a step-by-step plan to carry out
the user's request

Tools

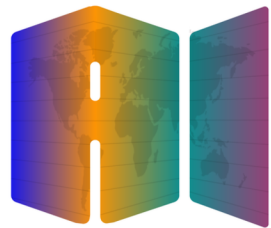
researcher

graphic
designer

writer

Create a summer
marketing
campaign for
sunglasses

LLM



Example: Planning with multiple agents

system
prompt

You have access to the following tools:

{description of tools}

Return a step-by-step plan to carry out
the user's request

Tools

researcher

graphic
designer

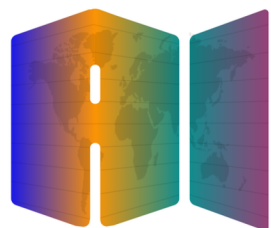
writer

Create a summer
marketing
campaign for
sunglasses

LLM

marketing
manager

1. Ask **researcher** to research current sunglasses trends
2. Ask **graphic designer** to create ad images
3. Ask **writer** to create report
4. Review report



Step 1 text

researcher

Step 1 output
Step 2 text

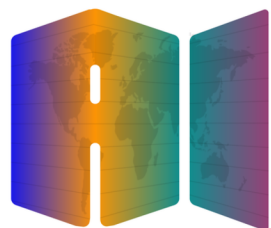
graphic
designer

Step 2 output
Step 3 text

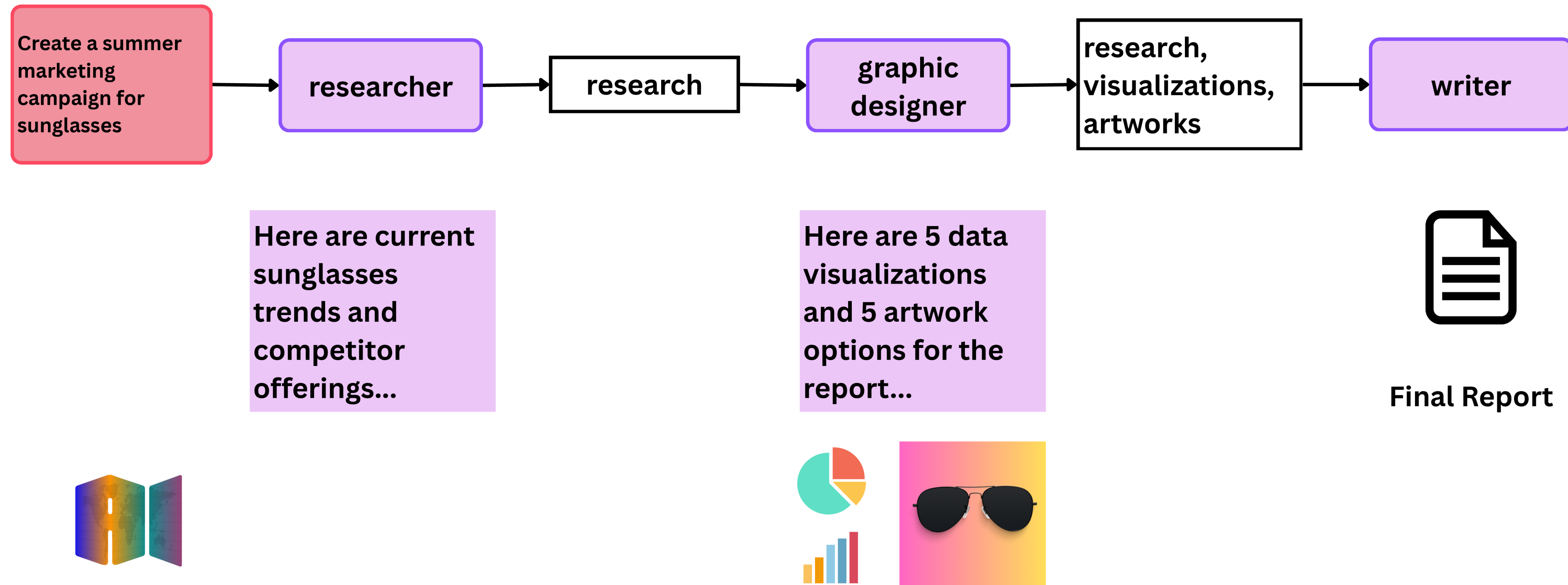
writer

Patterns for highly autonomous agents

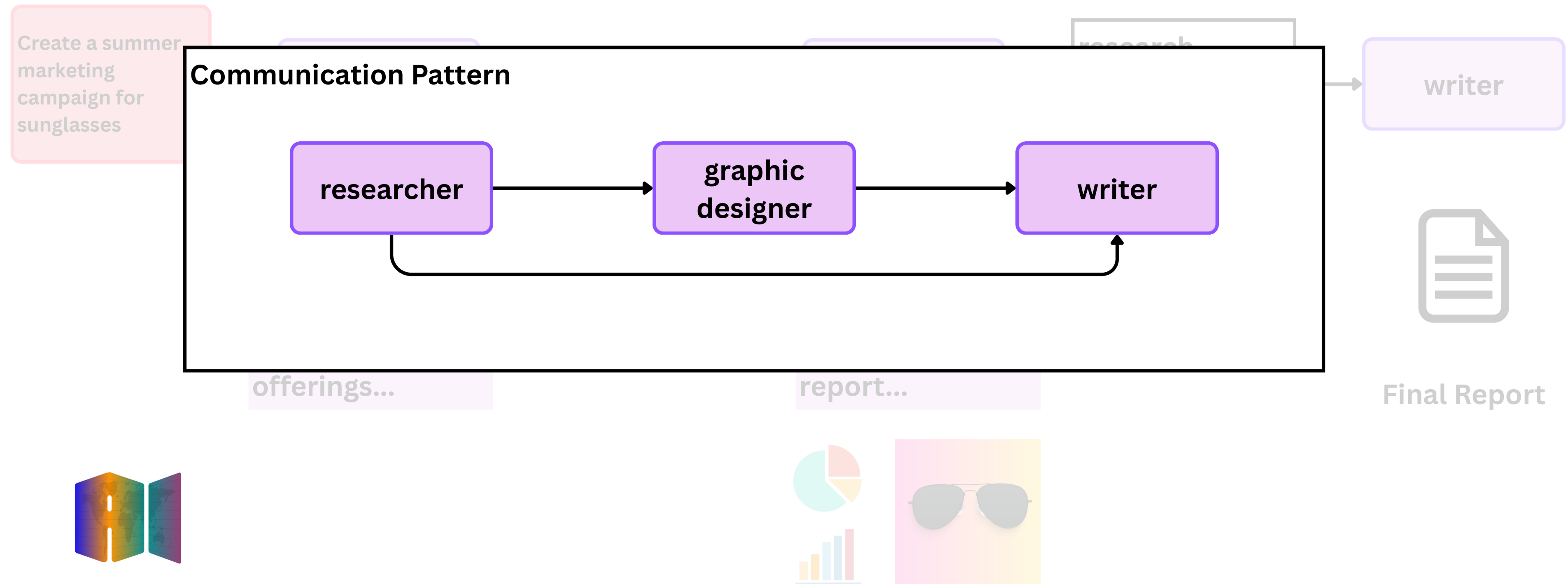
**Communication patterns
for multi-agent systems**



Example: Marketing team with linear plan



Example: Marketing team with linear plan



Example: Planning with multiple agents

system
prompt

You have access to the following tools:

{description of tools}

Return a step-by-step plan to carry out
the user's request

Tools

researcher

graphic
designer

writer

Create a summer
marketing
campaign for
sunglasses

LLM

marketing
manager

1. Ask **researcher** to research current sunglasses trends
2. Ask **graphic designer** to create ad images
3. Ask **writer** to create report
4. Review report

Step 1 text

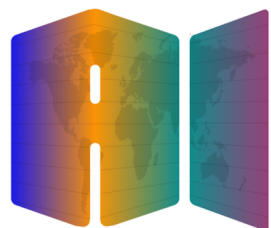
researcher

Step 1 output
Step 2 text

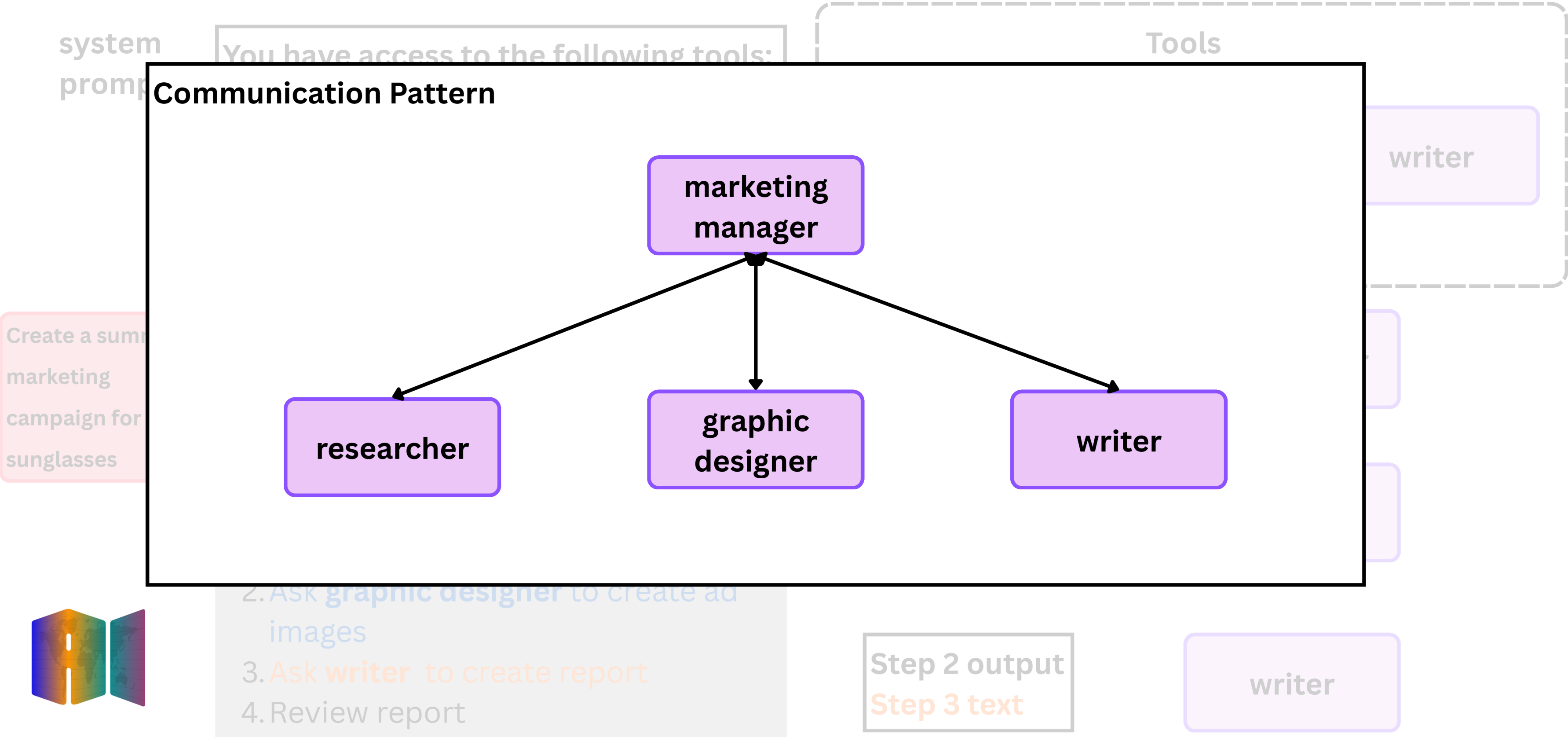
graphic
designer

Step 2 output
Step 3 text

writer

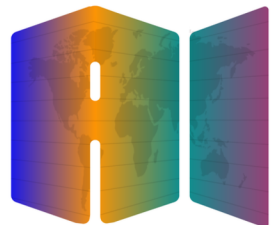
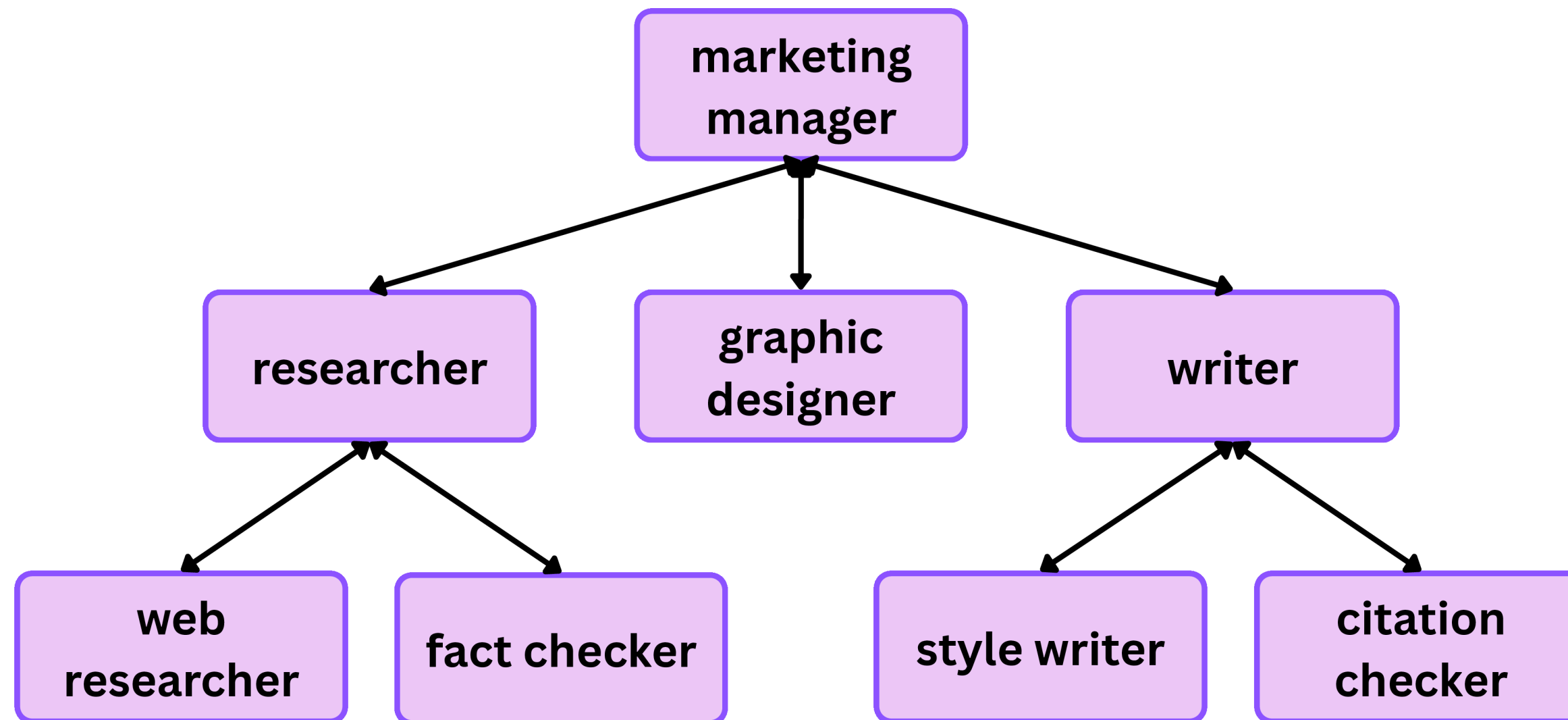


Example: Planning with multiple agents



Other communication patterns

Deeper hierarchy



All-to-all

